

טרנזיסטור בזרם חילופין

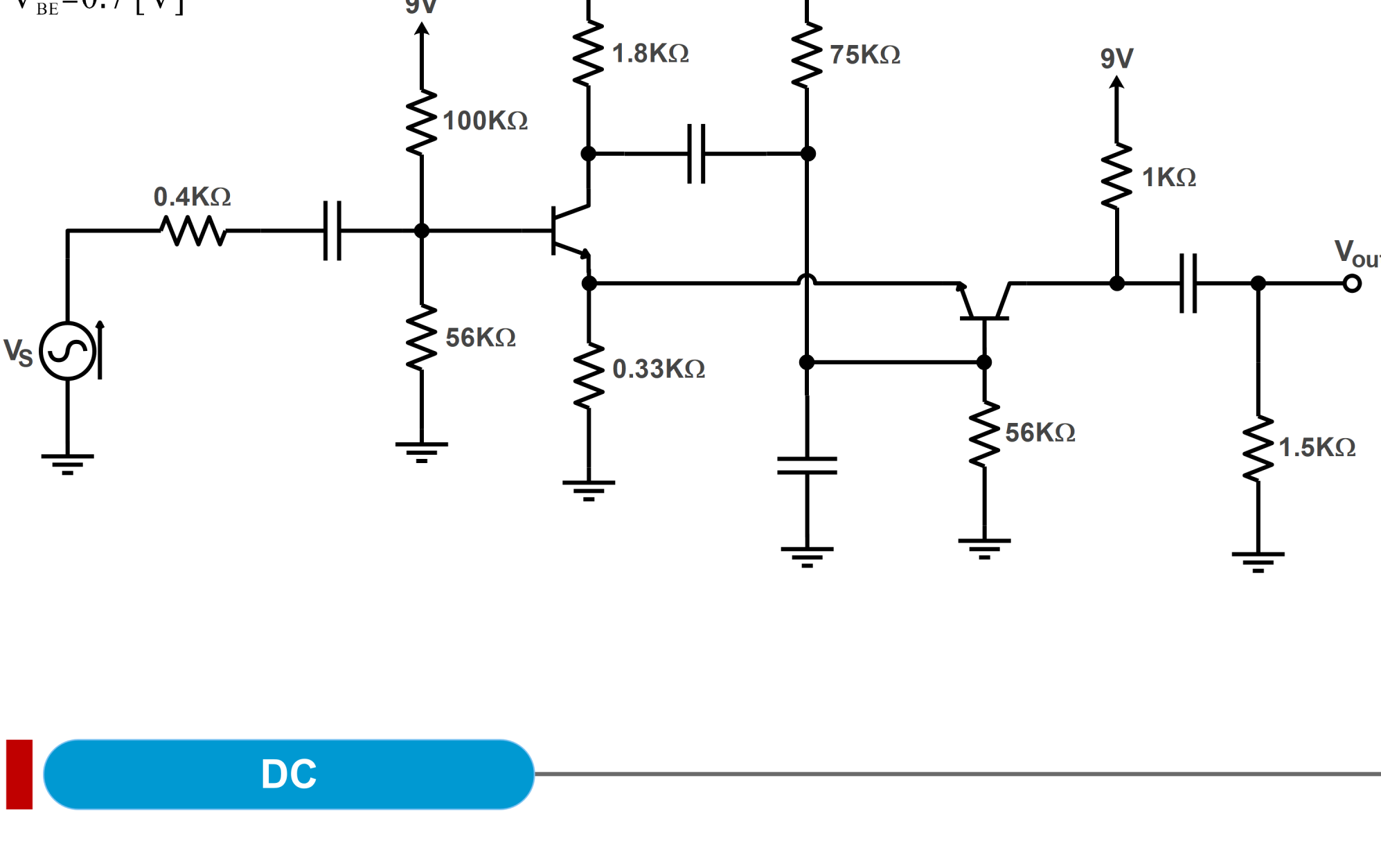
טרנזיסטור מסכס

$$\beta = h_{fe} = 50$$

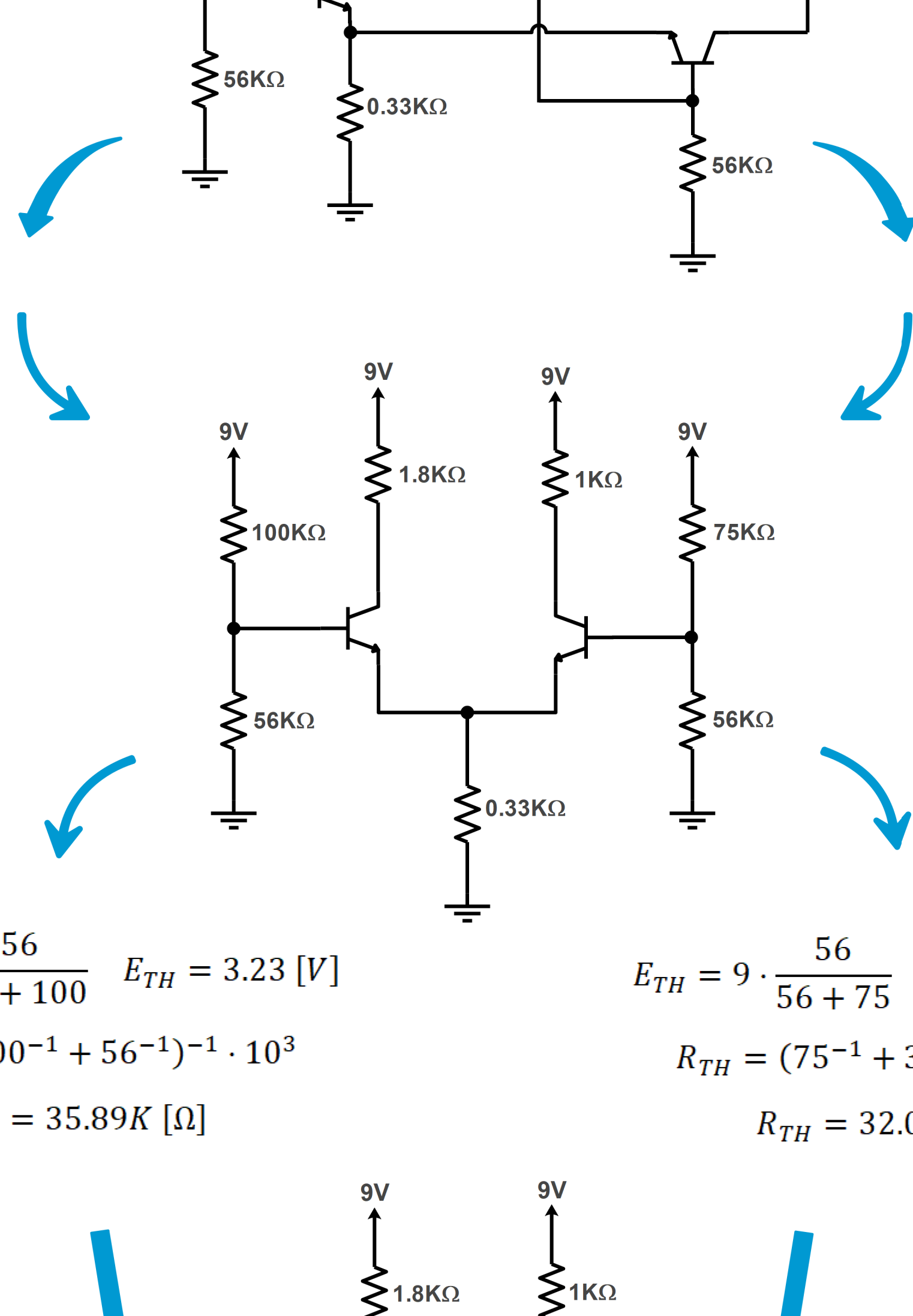
$$Q_1 = h_{ic} = 420 [\Omega]$$

$$Q_2 = h_{ic} = 820 [\Omega]$$

$$V_{BE} = 0.7 [V]$$



DC



$$E_{TH} = 9 \cdot \frac{56}{56 + 100} \quad E_{TH} = 3.23 [V]$$

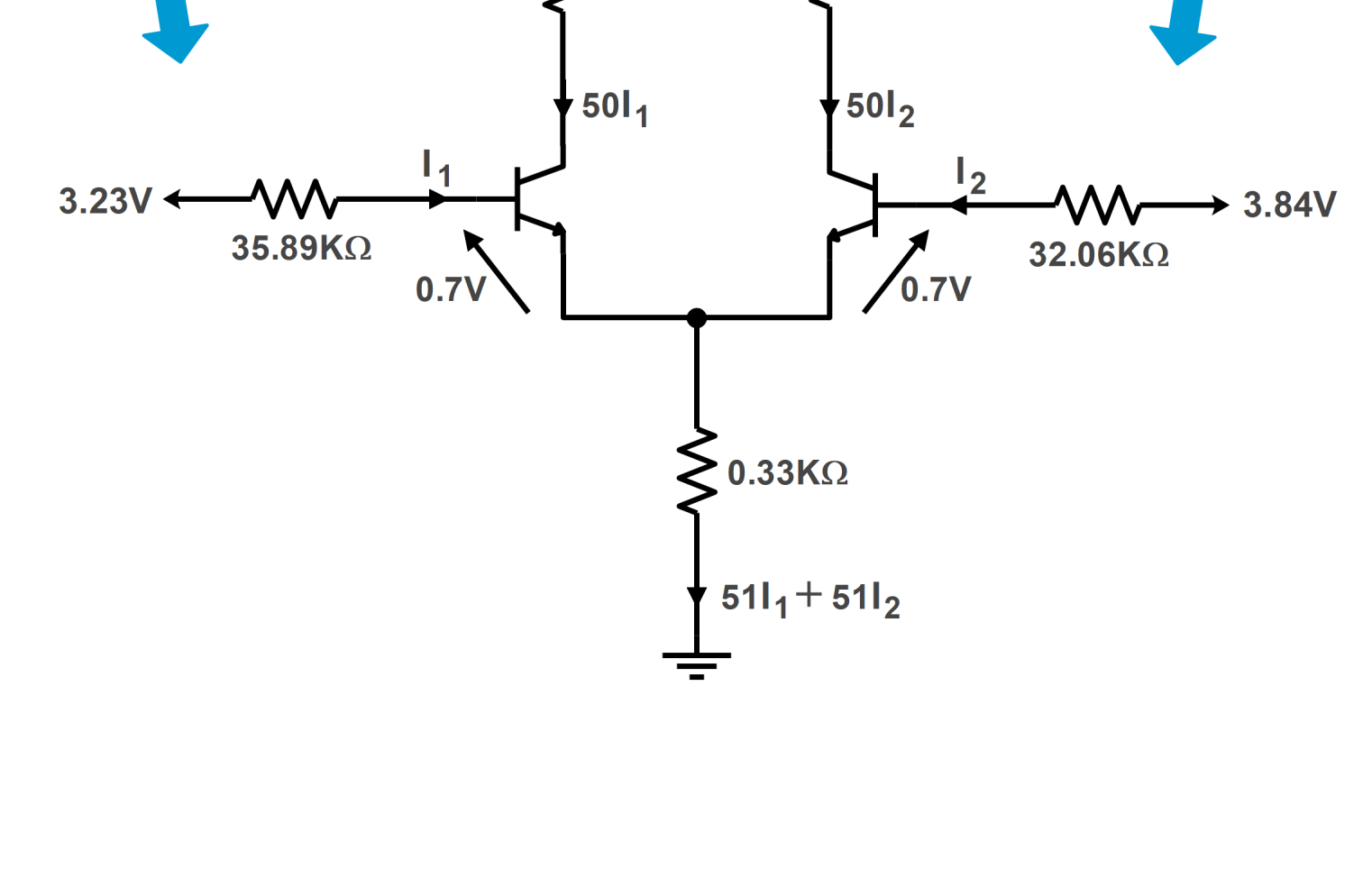
$$R_{TH} = (100^{-1} + 56^{-1})^{-1} \cdot 10^3$$

$$R_{TH} = 35.89K [\Omega]$$

$$E_{TH} = 9 \cdot \frac{56}{56 + 75} \quad E_{TH} = 3.84 [V]$$

$$R_{TH} = (75^{-1} + 36^{-1})^{-1} \cdot 10^3$$

$$R_{TH} = 32.06K [\Omega]$$



$$3.23 = 35.89 \cdot 10^3 \cdot I_1 + 0.7 + 16.83 \cdot 10^3 \cdot (I_1 + I_2)$$

$$2.53 = 52.72 \cdot 10^3 \cdot I_1 + 16.83 \cdot 10^3 \cdot I_2$$

$$I_2 = \frac{2.53 - 52.72 \cdot 10^3 \cdot I_1}{16.83 \cdot 10^3}$$

$$3.84 = 32.06 \cdot 10^3 \cdot I_2 + 0.7 + 16.83 \cdot 10^3 \cdot (I_1 + I_2)$$

$$3.14 = 32.06 \cdot 10^3 \cdot I_2 + 16.83 \cdot 10^3 \cdot I_1 + 16.83 \cdot 10^3 \cdot I_2$$

$$I_2 = \frac{3.14 - 16.83 \cdot 10^3 \cdot I_1}{48.89 \cdot 10^3}$$

$$\frac{2.53 - 52.72 \cdot 10^3 \cdot I_1}{16.83 \cdot 10^3} = \frac{3.14 - 16.83 \cdot 10^3 \cdot I_1}{48.89 \cdot 10^3}$$

$$2.9(2.52 - 52.72 \cdot 10^3 \cdot I_1) = 3.14 - 16.83 \cdot 10^3 \cdot I_1$$

$$4.2 - 153 \cdot 10^3 \cdot I_1 = -16.83 \cdot 10^3 \cdot I_1$$

$$4.2 = 136.34 \cdot 10^3 \cdot I_1$$

$$I_1 = 30.838 \mu [A]$$

$$I_2 = \frac{3.14 - 16.83 \cdot 10^3 \cdot 30.83 \cdot 10^{-6}}{48.89 \cdot 10^3} \Rightarrow I_2 = 53.75 \mu [A]$$

$$I_{C1} = 50 \cdot 30.83 \cdot 10^{-6} \quad I_{C1} = 1.54m [A]$$

$$I_{E1} = 51 \cdot 30.83 \cdot 10^{-6} \quad I_{E1} = 1.57m [A]$$

$$I_{C2} = 50 \cdot 53.75 \cdot 10^{-6} \quad I_{C2} = 2.68m [A]$$

$$I_{E2} = 51 \cdot 53.75 \cdot 10^{-6} \quad I_{E2} = 2.74m [A]$$

$$I_{RB1} = \frac{51(30.83 \cdot 10^{-6} + 53.75 \cdot 10^{-6}) \cdot 330 + 0.7}{56 \cdot 10^3} + 30.83 \cdot 10^{-6}$$

$$I_{RB1} = 68.76 \mu [A]$$

$$I_{RB2} = \frac{51(30.83 \cdot 10^{-6} + 53.75 \cdot 10^{-6}) \cdot 330 + 0.7}{56 \cdot 10^3} + 53.75 \cdot 10^{-6}$$

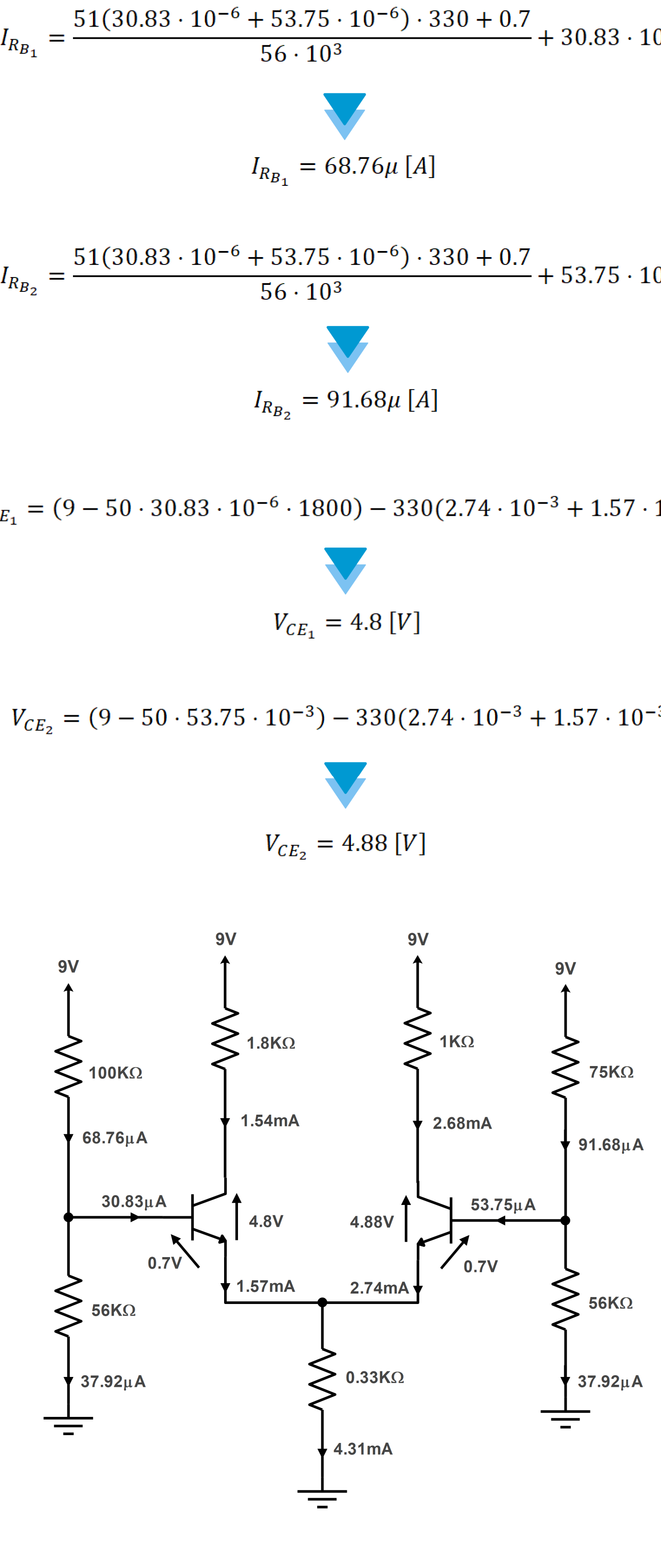
$$I_{RB2} = 91.68 \mu [A]$$

$$V_{CE1} = (9 - 50 \cdot 30.83 \cdot 10^{-6} \cdot 1800) - 330(2.74 \cdot 10^{-3} + 1.57 \cdot 10^{-3})$$

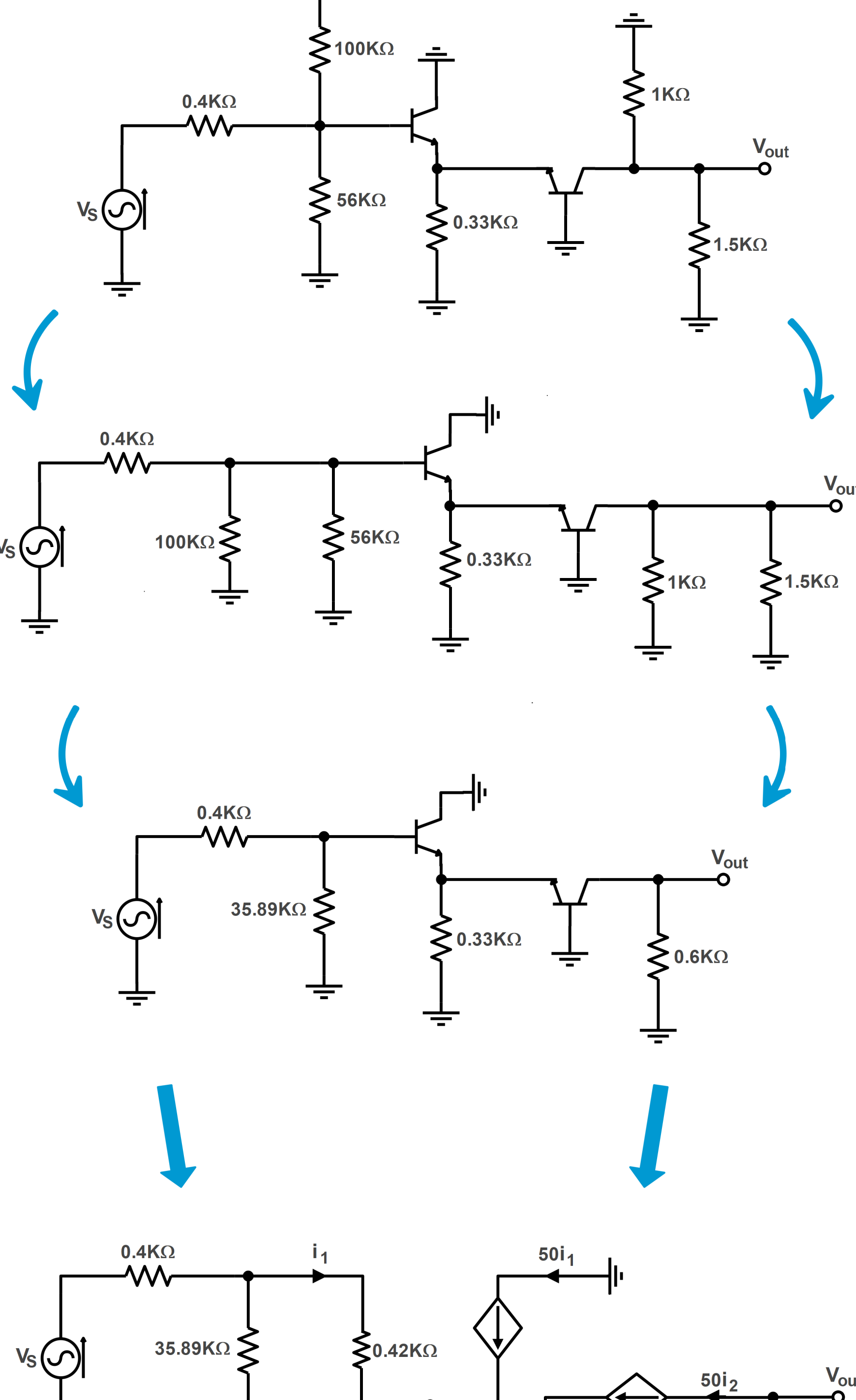
$$V_{CE1} = 4.8 [V]$$

$$V_{CE2} = (9 - 50 \cdot 53.75 \cdot 10^{-3}) - 330(2.74 \cdot 10^{-3} + 1.57 \cdot 10^{-3})$$

$$V_{CE2} = 4.88 [V]$$



AC



$$V_X = 420 \cdot i_1 + 16.83 \cdot 10^3 \cdot i_1 + 16.83 \cdot 10^3 \cdot i_2$$

$$V_X = 17.25 \cdot 10^3 \cdot i_1 + 16.83 \cdot 10^3 \cdot i_2$$

$$V_X = 420i_1 - 820i_2$$

$$420i_1 - 820i_2 = 17.25 \cdot 10^3 \cdot i_1 + 16.83 \cdot 10^3 \cdot i_2$$

$$-16.83 \cdot 10^3 \cdot i_1 = 17.65 \cdot 10^3 \cdot i_2$$

$$i_1 = -\frac{1765}{1683} \cdot i_2 \quad i_2 = -\frac{1683}{1765} \cdot i_1$$

$$V_X = 420 \left(-\frac{1765}{1683} \cdot i_2 \right) - 820i_2 \Rightarrow V_X = -1.26 \cdot 10^3 \cdot i_2$$

$$i_{in} = \frac{-1.26 \cdot 10^3 \cdot i_2}{35.89 \cdot 10^3} + \left(-\frac{1765}{1683} \cdot i_2 \right) \Rightarrow i_{in} = -1.08 \cdot i_2$$

$$V_S = 400 \cdot (-1.08 \cdot i_2) - 1.26 \cdot 10^3 \cdot i_2 \Rightarrow V_S = -1.69 \cdot 10^3 \cdot i_2$$

$$V_{out} = -(600 \cdot 50i_2)$$

$$\frac{V_{out}}{V_S} = \frac{-600 \cdot 50i_2}{-1.69 \cdot 10^3 \cdot i_2}$$

$$V_{out} = 17.71$$

$$V_X = V_S \cdot \frac{R_{in}}{R_{in} + 400} \Rightarrow V_X \cdot R_{in} + 400V_X = V_S \cdot R_{in} \quad V_X \cdot R_{in} + 400V_X - V_S \cdot R_{in} = 0$$

$$R_{in} \cdot (V_X - V_S) + 400V_X = 0 \Rightarrow R_{in} = -\frac{400V_X}{V_X - V_S}$$

$$R_{in} = -\frac{400 \cdot (-1.26 \cdot 10^3 \cdot i_2)}{(-1.26 \cdot 10^3 \cdot i_2) - (-1.69 \cdot 10^3 \cdot i_2)} \Rightarrow R_{in} = 1.16K [\Omega]$$

$$R_S = \frac{-1.69 \cdot 10^3 \cdot i_2}{-1.08 \cdot i_2} \Rightarrow R_S = 1.56K [\Omega]$$

$$\frac{i_{out}}{i_{in}} = 17.71 \cdot \frac{1.56 \cdot 10^3}{1.5 \cdot 10^3} \Rightarrow \frac{i_{out}}{i_{in}} = 18.45$$